

A-level

# ENVIRONMENTAL SCIENCE

## Questions by topic: Energy & material cycles: natural vs human systems - Set 1

### Mark scheme

AQA spec 3.6.2-3.6.3; Sustainability; textbook pp. 373-374

#### Marking guidance

- The number in brackets after each marking point is the mark it earns; the note above the points says how many are needed.
- Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

**Textbook references:** every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

| Question    | Marking guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Marks |
|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| <b>01.1</b> | <p><i>1 mark for each point.</i></p> <ul style="list-style-type: none"> <li>• Natural processes are driven by renewable, low energy-density solar energy captured by photosynthesis; human processes usually use high energy-density non-renewable fossil fuels <b>(1)</b></li> <li>• Natural processes occur at low temperatures because enzymes lower the activation energy of reactions; human processes need high temperatures and pressures <b>(1)</b></li> <li>• Natural processes produce no toxic wastes / their products are raw materials for other processes; industrial processes emit CO<sub>2</sub> and NO<sub>x</sub> <b>(1)</b></li> <li>• So natural fixation is sustainable indefinitely with a low carbon footprint <b>(1)</b></li> </ul> | 4     |
| <b>01.2</b> | <p><i>Any 3 from the points below - 1 mark each, maximum 3.</i></p> <ul style="list-style-type: none"> <li>• Hydrological cycle (evaporation, precipitation) <b>(1)</b></li> <li>• Carbon cycle / photosynthesis <b>(1)</b></li> <li>• Atmospheric circulation / winds <b>(1)</b></li> <li>• Thermohaline circulation (density differences from heating / evaporation) <b>(1)</b></li> <li>• Decomposition <b>(1)</b></li> </ul>                                                                                                                                                                                                                                                                                                                             | 3     |
| <b>01.3</b> | <p><i>1 mark for each point.</i></p> <ul style="list-style-type: none"> <li>• Bioleaching and phytomining extract metals at low temperature instead of high-temperature smelting <b>(1)</b></li> <li>• Anaerobic digestion and composting treat wastes; enzymes in industry (biological washing powders, biofuels from algae / microorganisms) <b>(1)</b></li> <li>• Heat pumps and solar thermal capture low-density heat; low-temperature manufacturing using biological polymers; fuel cells instead of combustion <b>(1)</b></li> </ul>                                                                                                                                                                                                                  | 3     |

| Question    | Marking guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Marks |
|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
|             | <b>Source:</b> R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 373; AQA spec 3.6.2.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |       |
| <b>02.1</b> | <p><i>1 mark for each point.</i></p> <ul style="list-style-type: none"> <li>Natural cycles use a small number of abundant elements (C, H, O, N, P) built into monomers and polymers (carbohydrates, proteins); wastes of one process are raw materials for another so nothing is depleted or accumulates <b>(1)</b></li> <li>Natural wastes are biodegradable / non-toxic and are broken down by decomposers into raw materials <b>(1)</b></li> <li>Human systems are linear: extract - process - manufacture - use - discard; minerals are dispersed after use or mixed so recovery is difficult <b>(1)</b></li> <li>Human wastes are often persistent and toxic (organochlorines, PCBs, heavy metals in electronics) and build up; fossil fuel use is inherently linear and unsustainable <b>(1)</b></li> </ul> | 4     |
| <b>02.2</b> | <p><i>1 mark for each point.</i></p> <ul style="list-style-type: none"> <li>Metals are dispersed in small amounts within complex mixed products so they cannot easily be separated / recovered at end of life <b>(1)</b></li> <li>They leach from landfill or are released by incineration; persistent, bioaccumulating <b>(1)</b></li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 2     |
| <b>02.3</b> | <p><i>1 mark for each point.</i></p> <ul style="list-style-type: none"> <li>Design for durability, repair and re-use (replaceable parts, no built-in batteries, spare parts and support, no planned obsolescence) <b>(1)</b></li> <li>Easy identification of materials (coding) and separation of components so they can be recycled <b>(1)</b></li> <li>Separation of technical materials (metals, plastics) from biological materials (which can be composted / digested) <b>(1)</b></li> <li>Use of recycled and recyclable / biodegradable materials; reduced packaging and material use (Cradle to Cradle) <b>(1)</b></li> </ul>                                                                                                                                                                             | 4     |
|             | <b>Source:</b> R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 373-374; AQA spec 3.6.3.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |       |
| <b>03.1</b> | <p><i>Any 2 from the points below - 1 mark each, maximum 2.</i></p> <ul style="list-style-type: none"> <li>Hydrological cycle <b>(1)</b></li> <li>Carbon cycle <b>(1)</b></li> <li>Nitrogen cycle <b>(1)</b></li> <li>Atmospheric circulation <b>(1)</b></li> <li>Thermohaline circulation <b>(1)</b></li> <li>Photosynthesis <b>(1)</b></li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 2     |
| <b>03.2</b> | <p><i>1 mark for each point.</i></p> <ul style="list-style-type: none"> <li>Enzymes reduce the activation energy of reactions <b>(1)</b></li> <li>So reactions such as photosynthesis, decomposition and nitrogen fixation do not need high temperatures <b>(1)</b></li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 2     |
| <b>03.3</b> | <p><i>Any 1 from the points below - 1 mark each, maximum 1.</i></p> <ul style="list-style-type: none"> <li>Haber process (nitrogen fixation) <b>(1)</b></li> <li>Incineration of wastes (decomposition) <b>(1)</b></li> <li>Many manufacturing processes <b>(1)</b></li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 1     |
|             | <b>Source:</b> R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 373; AQA spec 3.6.2.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |       |

| Question    | Marking guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Marks |
|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| <b>04.1</b> | <p><i>1 mark for each point.</i></p> <ul style="list-style-type: none"> <li>• Energy: low energy-density renewable (solar) / non-renewable fossil fuels (1)</li> <li>• Materials: cyclic - wastes of one process are raw materials of the next / linear - used once then discarded, depletion (1)</li> <li>• Wastes: biodegradable, non-toxic or do not build up / toxic, persistent (eg PCBs, organochlorines, toxic metals) that bioaccumulate (1)</li> <li>• Temperature: low / high (Haber process, incineration) (1)</li> </ul> | 4     |
| <b>04.2</b> | <p><i>1 mark for each point.</i></p> <ul style="list-style-type: none"> <li>• Abundant simple raw materials build into monomers and then a wide range of polymers (carbohydrates, proteins) (1)</li> <li>• They are re-used in cycles so there is no depletion of raw materials or build-up of wastes (1)</li> </ul>                                                                                                                                                                                                                 | 2     |
| <b>04.3</b> | <p><i>1 mark for each point.</i></p> <ul style="list-style-type: none"> <li>• Use often disperses the minerals (eg in products spread over the world / corrosion) (1)</li> <li>• Or produces mixtures of materials that are difficult to separate, so sustainable exploitation is reduced (1)</li> </ul>                                                                                                                                                                                                                             | 2     |
| <b>04.4</b> | <p><i>1 mark for each point.</i></p> <ul style="list-style-type: none"> <li>• They cannot easily be removed from end-of-life appliances and are not biodegradable (1)</li> <li>• Natural wastes are non-toxic / broken down to raw materials for other processes, whereas persistent toxins may bioaccumulate and biomagnify (1)</li> </ul>                                                                                                                                                                                          | 2     |
|             | <b>Source:</b> R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 373-374; AQA spec 3.6.2-3.6.3.                                                                                                                                                                                                                                                                                                                                                                                                          |       |
| <b>05.1</b> | <p><i>Correct answer only.</i></p> <ul style="list-style-type: none"> <li>• Waste products that are the raw materials for other processes (1)</li> </ul>                                                                                                                                                                                                                                                                                                                                                                             | 1     |
| <b>05.2</b> | <p><i>1 mark for each point.</i></p> <ul style="list-style-type: none"> <li>• Bacteria fix nitrogen at low temperature using enzymes and energy ultimately from sunlight (renewable) (1)</li> <li>• The Haber process needs high temperature and pressure powered by non-renewable fossil fuels, releasing CO<sub>2</sub> (1)</li> </ul>                                                                                                                                                                                             | 2     |
| <b>05.3</b> | <p><i>1 mark for each point.</i></p> <ul style="list-style-type: none"> <li>• Incineration uses high temperatures / fuel and releases pollutants; nutrients are not returned as usable raw materials (1)</li> <li>• Decomposition at low temperature recycles nutrients as the raw materials for other natural processes (material cycle) (1)</li> </ul>                                                                                                                                                                             | 2     |
|             | <b>Source:</b> R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 373; AQA spec 3.6.2-3.6.3.                                                                                                                                                                                                                                                                                                                                                                                                              |       |